

C.W

## Eigen Value and Eigen Vectors

□ Eigen Value: Let,  $A$  be any square matrix and  $\lambda$  be any scalar. Then,  $\lambda$  is called eigen value if there is any vector  $V$  which satisfies  $\lambda V = AV$ .

□ Eigen Vector: In  $\lambda V = AV$ ;  $V$  is called eigen vector corresponding to eigen values.

□ Characteristics Matrix:  $(A - \lambda I)$  or  $(\lambda I - A)$

here,

$A$  = square matrix

$\lambda$  = eigen value

$I$  = Identity / unit matrix

□ Characteristics equation: A matrix - 20-31

$$|A - \lambda I| = 0$$

## Caley Hamilton Theorem:

Statement: Every square matrix satisfies

its characteristics eqn.

(\*) Let  $A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 5 & 3 \\ 1 & 0 & 8 \end{pmatrix}$ ; verify Hamilton theorem

and find  $A^{-1}$ .

Solution:

Here,

$$|A - \lambda I| = 0$$

We know,

every square matrix satisfies its

characteristics eqn.

$$\text{i.e. } |A - \lambda I| = 0$$

$$\Rightarrow \left| \begin{pmatrix} 1 & 2 & 3 \\ 2 & 5 & 3 \\ 1 & 0 & 8 \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \right| = 0$$

$$\Rightarrow \left| \begin{pmatrix} 1 & 2 & 3 \\ 2 & 5 & 3 \\ 1 & 0 & 8 \end{pmatrix} - \begin{pmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{pmatrix} \right| = 0$$

$$\Rightarrow \begin{vmatrix} 1-\lambda & 2 & 3 \\ 2 & 5-\lambda & 3 \\ 1 & 0 & 8-\lambda \end{vmatrix} = 0$$

$$\Rightarrow (1-\lambda) \{ (5-\lambda)(8-\lambda) - 0 \} - 2 \{ 2(8-\lambda) - 3 \}$$

$$+ 3 \{ 0 + (5-\lambda) \} = 0 \quad ; \quad \begin{pmatrix} 8 & 5 & 1 \\ 2 & 5 & 0 \\ 3 & 0 & 1 \end{pmatrix} = A$$

$$\Rightarrow (1-\lambda)(40 - 13\lambda + \lambda^2) - 2(16 - 2\lambda - 3) - (15 - 3\lambda) = 0$$

$$\Rightarrow 40 - 13\lambda + \lambda^2 - 40\lambda + 13\lambda^2 - \lambda^3 - 26 + 4\lambda - 15 + 3\lambda = 0$$

$$\Rightarrow -\lambda^3 + 14\lambda^2 - 46\lambda - 1 = 0$$

$$\Rightarrow \lambda^3 - 14\lambda^2 + 46\lambda + 1 = 0$$

$$\Rightarrow \lambda = -0.02, 8.69, 5.32$$

$$\Rightarrow A^3$$

$$\Rightarrow A^3 - 14A^2 + 46A + 1 = 0$$

$$\begin{vmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{vmatrix} = 0$$

$$\begin{vmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{vmatrix} = 0$$

NOW,

$$A^2 = A \times A$$

$$= \begin{pmatrix} 1 & 2 & 3 \\ 2 & 5 & 3 \\ 1 & 0 & 8 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 \\ 2 & 5 & 3 \\ 1 & 0 & 8 \end{pmatrix}$$

$$= \begin{pmatrix} 1+4+3 & 2+10+0 & 3+6+24 \\ 2+10+3 & 4+25+0 & 6+15+24 \\ 1+0+8 & 2+0+0 & 3+0+64 \end{pmatrix}$$

$$= \begin{pmatrix} 8 & 12 & 33 \\ 15 & 29 & 45 \\ 9 & 2 & 67 \end{pmatrix}$$

$$\therefore A^3 = A^2 \times A$$

$$= \begin{pmatrix} 8 & 12 & 33 \\ 15 & 29 & 45 \\ 9 & 2 & 67 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 \\ 2 & 5 & 3 \\ 1 & 0 & 8 \end{pmatrix}$$

$$= \begin{pmatrix} 8+24+33 & 16+60+0 & 24+36+264 \\ 15+58+45 & 30+145+0 & 45+87+536 \\ 9+4+67 & 18+10+0 & 27+6+536 \end{pmatrix}$$

$$= \begin{pmatrix} 65 & 76 & 324 \\ 118 & 175 & 668 \\ 80 & 28 & 569 \end{pmatrix}$$

... ..

$$\therefore A^3 - 14A^2 + 46A + I$$

$$= \begin{bmatrix} 65 & 76 & 324 \\ 118 & 175 & 492 \\ 80 & 28 & 569 \end{bmatrix} - 14 \begin{bmatrix} 8 & 12 & 33 \\ 15 & 29 & 45 \\ 9 & 2 & 67 \end{bmatrix}$$

$$+ 46 \begin{bmatrix} 12 & 3 \\ 25 & 3 \\ 10 & 8 \end{bmatrix}$$

$$+ \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 65 & 76 & 324 \\ 118 & 175 & 492 \\ 80 & 28 & 569 \end{bmatrix} - \begin{bmatrix} 112 & 168 & 462 \\ 210 & 406 & 630 \\ 126 & 28 & 938 \end{bmatrix}$$

$$+ \begin{bmatrix} 46 & 92 & 138 \\ 92 & 230 & 138 \\ 46 & 0 & 368 \end{bmatrix} + \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 65 - 112 + 46 + 1 & 76 - 168 + 92 + 0 & 324 - 462 + 138 + 0 \\ 118 - 210 + 92 + 0 & 175 - 406 + 230 + 1 & 492 - 630 + 138 + 0 \\ 80 - 126 + 46 + 0 & 28 - 28 + 0 + 0 & 569 - 938 + 368 + 1 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = O$$

So, Cayley Hamilton theorem is verified.

HW: Verify Cayley Hamilton theorem for

$$A = \begin{bmatrix} 2 & 2 & 1 \\ 1 & 3 & 1 \\ 1 & 2 & 2 \end{bmatrix}$$

Q1. W  $\{x=0\} = \{x=0\} = \{x=0\} = \{x=0\}$  29.09.2026

(\*) Verify Cayley Hamilton theorem for

$$A = \begin{bmatrix} 2 & 2 & 1 \\ 1 & 3 & 1 \\ 1 & 2 & 2 \end{bmatrix}$$

Solution:

We know that, every square matrix satisfies